Local well-posedness for transport-noise Navier–Stokes with $B^{-1/2}_{6,4}(\mathbb{R}^3)$ initial data

Kaiwen Jin
yui.ui.siki@gmail.com

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Abstract

We prove local well-posedness of the three-dimensional incompressible Navier–Stokes equation with prescribed smooth, compactly supported, divergence-free Ito transport noise and deterministic initial velocity in the inhomogeneous space $B^{-1/2}_{6,4}(\mathbb{R}^3)$. Under strict stochastic parabolicity, with no further smallness assumption, the solution is heat-mild and belongs almost surely to $C_t B^{-1/2}_{6,4} \cap L^4_t L^6_x$ up to a positive stopping time. Pathwise uniqueness holds in this entire class, and initial-data dependence is continuous in probability in both norms on common local intervals. The proof combines the published linear stochastic Stokes maximal-regularity theorem of Agresti and Veraar with a cutoff depending on the accumulated $L^4_t L^6_x$ norm. This gives a nonlinear local theory at the negative-half endpoint without treating the derivative noise as a small perturbation.

1 Introduction and main result

The transport-noise Navier–Stokes equation replaces part of the advecting velocity by random spatial fields. After projecting onto incompressible velocities, the Ito equation on $\mathbb{R}^3$ takes the form

$$du = [\nu\Delta u - \mathbb{P}\,\mathrm{div}(u\otimes u)]\,dt + \sum_{k=1}^{K} \mathbb{P}[(b_k\cdot\nabla)u]\,dW^k_t. \tag{1.1}$$

Here $\nu > 0$ is viscosity, $\mathbb{P}$ is the Helmholtz projection, and W^k are independent standard real Brownian motions. We consider real fields $b_k \in C^\infty_c(\mathbb{R}^3;\mathbb{R}^3)$ with $\mathrm{div}\, b_k = 0$ and assume that, for some $\delta > 0$,

$$\nu I - \frac{1}{2}\sum_{k=1}^{K} b_k(x)\otimes b_k(x) \geq \delta I \qquad (x\in\mathbb{R}^3). \tag{1.2}$$

This is the strict stochastic parabolicity condition. It permits the first-order noise to enter the principal linear equation, with constants depending on the prescribed fields.

Rough initial velocities require a solution theory compatible with viscous smoothing. The Navier–Stokes scaling assigns spatial smoothness $-1+3/q$ at integrability q. At $q=6$ this is $-1/2$, while the corresponding space-time norm is $L^4_tL^6_x$. These are high-frequency critical indices; the Besov space used here is inhomogeneous. Agresti and Veraar developed stochastic Stokes maximal regularity and critical-space nonlinear theory for transport-noise equations [1]. Their nonlinear theorem covers spatial integrability below six at the lowest smoothness. The endpoint is part of the rough-data question discussed in [2, Theorem 8.26 and Open Problem 10]. The linear theorem itself allows the Sobolev shift needed at $q=6$. Its nonlinear conclusion cannot simply be evaluated there: the critical time weight reaches the excluded

boundary of the stochastic maximal-regularity range. We use that linear theorem and a separate heat-mild quadratic estimate to obtain the endpoint.

Write $X = B^{-1/2}_{6,4,\sigma}(\mathbb{R}^3)$, with σ denoting divergence-free distributions. Work on any complete right-continuous filtered probability space carrying the Brownian motions above. By an adapted heat-mild solution on $[0,\tau]$ we mean an adapted continuous X process, with a progressive L^6 representative at almost every positive time, satisfying

$$u(t) = e^{\nu t\Delta}u_0 - \int_0^t e^{\nu(t-s)\Delta}\mathbb{P}\operatorname{div}(u\otimes u)(s)\,ds + \sum_{k=1}^{K}\int_0^t e^{\nu(t-s)\Delta}\mathbb{P}[(b_k\cdot\nabla)u(s)]\,dW_s^k, \qquad 0\le t\le\tau. \tag{1.3}$$

The identity is in distributions, equivalently in $H^{-2,6}$, with the continuous versions of the convolutions. Stochastic integrals are first defined at deterministic times after localization and then restricted to the stopped interval. This specifies the heat semigroup in both convolutions.

Theorem 1.1. *Let $\nu > 0$ and let the finite family $(b_k)_{k=1}^K$ satisfy the assumptions above, including* (1.2)*. For every deterministic $u_0 \in X$ there exist a stopping time $0 < \tau \le 1$ almost surely and an adapted heat-mild solution of* (1.1) *such that*

$$u \in C([0,\tau];X)\cap L^4(0,\tau;L^6(\mathbb{R}^3)) \quad \textit{almost surely.}$$

Any two heat-mild solutions in this class with the same initial value and Brownian motions agree on their common lifetime.

If deterministic $u_{0,n}\to u_0$ in X, the solutions can be chosen on the same Brownian space with positive lifetimes $\tau_n,\tau\le 1$ so that

$$\|u_n-u\|_{C([0,\tau_n\wedge\tau];X)} + \|u_n-u\|_{L^4(0,\tau_n\wedge\tau;L^6)} \longrightarrow 0 \quad \textit{in probability.}$$

There is also a stopping time $0<\rho\le\tau$ almost surely with $\Pr(\tau_n<\rho)\to 0$. No extra smallness of u_0 or of the fields is assumed.

The result addresses the specified $B^{-1/2}_{6,4}$ case of the rough-data question, for all the prescribed compactly supported fields satisfying (1.2). It does not assert global existence or determine the largest possible critical initial space.

The proof treats the full linear derivative noise before the quadratic term. The published linear estimate yields a bounded causal inverse for the identity minus the heat stochastic-convolution operator on $L^4(\Omega;L^4_tL^6_x)$. A cutoff depending on the past accumulated $L^4_tL^6_x$ norm then makes the drift globally Lipschitz with an arbitrarily small constant. Stopping before the cutoff acts recovers the original equation. For uniqueness among solutions with only almost-sure norms, we use bounded-norm random slabs and the stopped linear estimate.

Section 2 proves the heat estimates and specifies the published linear input, including the stopping-time inverse. Section 3 constructs the local solution and proves uniqueness and stability. We use $\widehat{f}(\xi)=\int e^{-ix\cdot\xi}f(x)\,dx$; $\mathbb{P}$ has symbol $I-\xi\otimes\xi/|\xi|^2$. All Besov and Sobolev spaces are inhomogeneous, and $\|\cdot\|_q$ denotes the spatial L^q norm.

2 Heat estimates and the linear stochastic equation

All spaces below are on $\mathbb{R}^3$, with vector or tensor values as appropriate. Write $J=(1-\Delta)^{1/2}$ and $H^{s,q}=\{f : J^sf\in L^q\}$. Fix an inhomogeneous smooth dyadic partition $(\Delta_j)_{j\ge -1}$, with one low-frequency block and annuli of size 2^j for $j\ge 0$. We use the norm

$$\|f\|_X = \Big(\sum_{j\ge -1} 2^{-2j}\|\Delta_j f\|_6^4\Big)^{1/4}, \qquad X = B^{-1/2}_{6,4,\sigma}. \tag{2.1}$$

The subscript σ denotes the distributional kernel of divergence. Different admissible partitions give equivalent norms by finite overlap and uniform L^1 bounds for their convolution kernels. The space X is a separable Banach space and embeds continuously into $H^{-2,6}_\sigma$. Indeed, if $a_j = 2^{-j/2}\|\Delta_j f\|_6$, the high-frequency contribution to the latter norm is bounded by $C\sum_{j\ge 0} 2^{-3j/2}a_j \le C\|(a_j)\|_{\ell^4}$. Completeness follows by compatible dyadic block limits, and separability by the closed embedding into $\ell^4(L^6)$.

For $0 < T < \infty$ set

$$E_T = L^4(0,T;L^6_\sigma), \qquad \mathcal{E}_T = L^4(\Omega;E_T)_{\mathrm{prog}}.$$

Here the second space consists of progressively measurable processes. It is a closed Banach subspace of the indicated Bochner space. For a stopping time $\gamma \le T$, $\|v\|_{E_\gamma}$ means $(\int_0^\gamma \|v(s)\|_6^4\,ds)^{1/4}$. Constants can depend on T, ν, the fixed fields and the parabolicity margin; no uniformity over coefficient families is asserted.

2.1 Deterministic heat estimates

Let $S(t) = e^{\nu t\Delta}$ and, for tensor-valued F, define

$$(DF)(t) = \int_0^t S(t-s)\mathbb{P}\operatorname{div} F(s)\,ds.$$

The following estimates include the time-continuous trace that will be needed for the stopping construction.

Lemma 2.1. *For each finite T there are finite constants such that*

$$\|DF\|_{E_T} \le C_D\|F\|_{L^2(0,T;L^3)}, \tag{2.2}$$

$$\|DF\|_{C([0,T];X)} \le C_X(T)\|F\|_{L^2(0,T;L^3)}, \qquad DF(0) = 0, \tag{2.3}$$

$$\|S(\cdot)f\|_{E_T} + \|S(\cdot)f\|_{C([0,T];X)} \le C_0(T)\|f\|_X. \tag{2.4}$$

Moreover $\mathbb{P}\operatorname{div} : L^3 \to H^{-2,6}_\sigma$ is bounded. The constant C_D can be chosen independently of T.

Proof. The Gaussian derivative bound, Young's inequality and boundedness of $\mathbb{P}$ on L^6 give

$$\|S(t)\mathbb{P}\operatorname{div} F\|_6 \le Ct^{-3/4}\|F\|_3. \tag{2.5}$$

The derivative contributes $t^{-1/2}$ and the L^3 to L^6 estimate contributes $t^{-1/4}$. Extend $\|F(t)\|_3$ by zero outside $(0,T)$. The one-dimensional Hardy–Littlewood–Sobolev inequality with fractional order $1/4$ maps $L^2(\mathbb{R})$ to $L^4(\mathbb{R})$ and proves (2.2). We use the finite-exponent forms of these inequalities; see [3, Corollary 6.3 and Theorem 6.4].

For the trace, choose enlarged blocks $\widetilde{\Delta}_j$ equal to one on the support of Δ_j. The localized heat kernel satisfies, for $j \ge 0$,

$$\|\Delta_j DF(t)\|_6 \le C2^{3j/2}\int_0^t e^{-c\nu 2^{2j}(t-s)}\|\widetilde{\Delta}_j F(s)\|_3\,ds.$$

To obtain this bound, scale the multiplier to a fixed annulus. Its smooth kernel has $L^{6/5}$ norm at most $C2^{3j/2}e^{-c\nu 2^{2j}(t-s)}$: the derivative costs 2^j, the spatial integrability change costs $2^{j/2}$, and integration by parts absorbs the polynomial factors in $\nu 2^{2j}(t-s)$ into the exponential. Cauchy–Schwarz in time therefore yields

$$\sup_{t\le T} 2^{-j/2}\|\Delta_j DF(t)\|_6 \le C\|\widetilde{\Delta}_j F\|_{L^2(0,T;L^3)}. \tag{2.6}$$

The low block is bounded by $C\sqrt{T}\|\widetilde{\Delta}_{-1}F\|_{L^2L^3}$, using its fixed Fourier support and heat contraction. For the high blocks, mixed-norm interchange and the upper Littlewood–Paley estimate give

$$\begin{aligned}\Big\|\big(\|\widetilde{\Delta}_j F\|_{L^2L^3}\big)_{j\ge0}\Big\|_{\ell^4} &\le \Big\|\big(\|\widetilde{\Delta}_j F\|_{L^2L^3}\big)_{j\ge0}\Big\|_{\ell^3}\\ &\le \Big\|\Big(\sum_{j\ge0}|\widetilde{\Delta}_j F|^3\Big)^{1/3}\Big\|_{L^2L^3}\\ &\le \Big\|\Big(\sum_{j\ge0}|\widetilde{\Delta}_j F|^2\Big)^{1/2}\Big\|_{L^2L^3} \le C\|F\|_{L^2L^3}.\end{aligned}$$

The mixed-norm direction uses $3 \ge 2$; the last estimate is the upper square-function bound at spatial exponent 3 [3, Proposition 4.1], [4, Proposition 5.3]. Together with (2.6), this proves the uniform X bound. Approximation by smooth functions in L^2L^3 gives continuity into X and zero initial trace, proving (2.3).

For (2.4), first let $T = 1$. The low block is in L^6. For the high blocks put $a_j = 2^{-j/2}\|\Delta_j f\|_6$. On $2^{-2(n+1)} < t \le 2^{-2n}$, $n \ge 0$, the annular heat bound implies

$$2^{-n/2}\left\|S(t)\sum_{j\ge0}\Delta_j f\right\|_6 \le C\sum_{j\ge0}2^{(j-n)/2}e^{-c\nu 2^{2(j-n-1)}}a_j.$$

The sequence $2^{\ell/2}e^{-c\nu 2^{2(\ell-1)}}$ lies in $\ell^1(\mathbb{Z})$. Integration over these time intervals and discrete Young's inequality give the E_1 bound. For continuity in X at zero, each $(S(t)-I)\Delta_j f$ tends to zero in L^6 and is bounded by $2\|\Delta_j f\|_6$. Dominated convergence in (2.1) proves the claim. Commutation with the blocks gives the uniform X bound and continuity at later times. If $T > 1$, heat contraction after time 1 gives finite T-dependent constants.

Finally the heat representation of J^{-2} and (2.5) with viscosity one give

$$\|\mathbb{P}\operatorname{div} F\|_{H^{-2,6}} \le C\int_0^\infty e^{-s}s^{-3/4}\,ds\,\|F\|_3 \le C\|F\|_3.$$

□

2.2 The published linear estimate

The tensor convention is $(v\otimes b)_{ij} = v_i b_j$ and $(\operatorname{div} F)_i = \sum_j \partial_j F_{ij}$. Thus $\operatorname{div} b_k = 0$ implies

$$B_k v := \mathbb{P}[(b_k\cdot\nabla)v] = \mathbb{P}\operatorname{div}(v\otimes b_k), \qquad \|Bv\|_{H^{-1,6}(\ell^2)} \le C_b\|v\|_6. \tag{2.7}$$

Here $Bv = (B_k v)_{k=1}^K$, and the bound follows from the bounded multiplier $J^{-1}\partial_j$ on L^6 and boundedness of the fields. The projection remains outside the variable coefficient derivative.

Proposition 2.2 (Agresti–Veraar, specialized)**.** *For every finite T and progressive $g \in L^4(\Omega; L^4(0,T; H^{-1,6}_\sigma(\ell^2)))$, the equation*

$$dz = \nu\Delta z\,dt + (Bz+g)\,dW, \qquad z(0) = 0, \tag{2.8}$$

has a unique strong solution with

$$\|z\|_{\mathcal{E}_T} + \|z\|_{L^4(\Omega;C([0,T];X))} \le C_L(T)\|g\|_{L^4(\Omega;L^4(0,T;H^{-1,6}(\ell^2)))}. \tag{2.9}$$

The same assertion holds with $B = 0$. The strong identity is in $H^{-2,6}$. It admits stopping-time restriction and pathwise-integrability localization; in particular, two local strong solutions with the same forcing and zero initial value agree before a common stopping time whenever their E norms are almost surely finite.

Source and parameter verification. Use [1, Assumption 3.1, Theorem 3.2 and Remark 3.3] with

$$(p, q, \kappa, \eta) = (4, 6, 0, -1), \qquad X_0 = H^{-2,6}_\sigma, \quad X_1 = L^6_\sigma, \quad X_{1/2} = H^{-1,6}_\sigma.$$

Here η denotes the spatial shift called δ in that source; it is unrelated to the positive margin δ in (1.2). The source linear assumption allows every real spatial shift. Its trace is

$$(X_0, X_1)_{3/4,4} = B^{-1/2}_{6,4,\sigma} = X, \qquad 1 + \eta - 2(1 + \kappa)/p = -1/2.$$

Take $a^{ij} = \nu\delta_{ij}$ and the source coefficients $c = h = 0$. The additional pressure-dependent drift in its general Stokes system then vanishes. The noise operator is exactly $\mathbb{P}(b_k \cdot \nabla)$. The source coefficient Sobolev regularity may be chosen of order 2 and integrability 6: $2 > 1 = -\eta$ and $2 > 3/6$. Smooth compactly supported b_k satisfy this condition. For the whole-space version, Remark 3.3 permits a constant background at infinity: $a = \nu I$ is that background and $b = 0$ outside a compact set. No global L^6 integrability of the nonzero constant matrix is needed. Its strict stochastic parabolicity condition is precisely (1.2). Extend the finite noise family by zeros in ℓ^2.

Theorem 3.2 now gives (2.9), including its continuous trace. The stopped and localized statement is the corresponding stochastic maximal regularity assertion: the proof of that theorem establishes the $\mathrm{SMR}^\bullet_{p,\kappa}$ property, and its Lemma 3.7 gives the stopped a priori estimate. Apply the pathwise-data localization in [2, Definitions 3.7–3.8 and Proposition 3.11] to this couple. This gives the stated local strong uniqueness as well as the deterministic interval estimate. These are linear results; no nonlinear endpoint theorem is being imported. □

2.3 Heat stochastic convolutions and a causal inverse

For $v \in \mathcal{E}_T$ define, initially at deterministic times,

$$(Kv)(t) = \sum_{k=1}^{K} \int_0^t S(t - s)B_k v(s)\, dW^k_s. \tag{2.10}$$

The integral is well defined in $H^{-1,6}$, since (2.7) and the finite-horizon embedding $L^4_t \subset L^2_t$ apply. The usual Banach-space Ito integration on these Sobolev spaces is available; see [2, Section 2.5 and Proposition 2.4]. The heat-only case of Proposition 2.2 shows that K has a continuous X version and

$$\|Kv\|_{\mathcal{E}_T} + \|Kv\|_{L^4(\Omega;C([0,T];X))} \le C_K(T)\|v\|_{\mathcal{E}_T}, \qquad Kv(0) = 0. \tag{2.11}$$

For clarity, the strong and heat-mild identities agree at this regularity. If Z solves the heat-only equation with forcing $g = Bv$, pair it, at time s, with $S(t - s)\phi$ for a Schwartz field ϕ and deterministic t. The derivative of the test field is $-\nu\Delta S(t - s)\phi$, so the Laplacian pairings cancel and

$$\langle Z(t), \phi\rangle = \sum_k \int_0^t \langle B_k v(s), S(t - s)\phi\rangle\, dW^k_s. \tag{2.12}$$

This testing rule follows from deterministic time partitions and summation by parts. Drift errors converge in $L^1_t H^{-2,6}$ and stochastic errors by the scalar Ito isometry after localization. Thus (2.12) identifies Z with (2.10), proving (2.11). A countable dense set of tests and rational times identifies their continuous versions. The same argument for (2.8) gives $z = Kz + K_g$, with K_g the heat stochastic convolution of g.

These identities also hold after stopping the cumulative E norm, for processes whose E norm is only almost surely finite. The deterministic terms satisfy the corresponding strong identities in $H^{-2,6}$: $\Delta S(\cdot)f \in L^4_t H^{-2,6}$ and, by Lemma 2.1, $\Delta DF \in L^4_t H^{-2,6}$ and

$\mathbb{P}\,\mathrm{div}\,F \in L^2_t H^{-2,6}$. Consequently a heat-mild solution in $E \cap CX$ is a strong $H^{-2,6}$ solution. The nonlinear drift is only required to be $L^1_t H^{-2,6}$; it is not used as L^4 drift forcing in Proposition 2.2. Evaluation of the continuous convolution at a stopping time means restriction of this version, not an Ito integral with a random terminal-time kernel.

Lemma 2.3. *On $\mathcal{E}_T$, $I-K$ has a bounded linear causal inverse R_T. There is a finite $M_T \geq 1$ such that, for every stopping time $\gamma \leq T$ and progressive w, v satisfying $w - Kw = v$ on $[0,\gamma]$ with finite $L^4(\Omega; E_\gamma)$ norms,*

$$\|w\|_{L^4(\Omega;E_\gamma)} \leq M_T \|v\|_{L^4(\Omega;E_\gamma)}. \tag{2.13}$$

Proof. Given $v \in \mathcal{E}_T$, solve (2.8) with $g = Bv$ and put $R_T v = v + z$. Then $\|R_T v\|_{\mathcal{E}_T} \leq (1 + C_L(T)C_b)\|v\|_{\mathcal{E}_T}$. The identity $z = Kz + Kv$ gives $(I-K)R_T v = v$. Conversely, if $w - Kw = v$, then $z = w - v = Kw$ is the zero-initial strong solution of (2.8) with forcing Bv, because $Bz + Bv = Bw$. Linear uniqueness gives $w = R_T v$. Equal inputs through a stopping time give equal forcing there; local linear uniqueness proves causality.

For the stopped estimate, $z = w - v = Kw$ solves (2.8) up to γ with $g = Bv$. Extend *this forcing* by zero after γ and solve the linear equation on $[0,T]$. Local uniqueness in Proposition 2.2 identifies its restriction with z. Restrict (2.9) and add v to obtain (2.13) with $M_T = 1 + C_L(T)C_b$. The solution itself is not extended by zero, so no jump at γ is introduced. This construction does not require $\|K\| < 1$ or a bounded pathwise inverse. □

3 Local solutions, uniqueness and stability

3.1 A cutoff depending on the past

For a path $u \in E_T$ let

$$A_u(t) = \Big(\int_0^t \|u(s)\|_6^4 \, ds \Big)^{1/4}.$$

It is continuous, nondecreasing, starts at zero, and satisfies $|A_u(t) - A_v(t)| \leq \|u - v\|_{E_t}$. For a progressive process it is adapted. Fix a nonincreasing Lipschitz function $\theta : [0,\infty) \to [0,1]$ with $\theta = 1$ on $[0,1]$ and $\theta = 0$ on $[2,\infty)$. Write L_θ for its Lipschitz constant. To truncate the quadratic drift without using future values, set

$$F_r(u)(t) = \theta(A_u(t)/r)\, u(t) \otimes u(t), \qquad r > 0.$$

Lemma 3.1. *For all $u, v \in E_T$,*

$$\|F_r(u)\|_{L^2L^3} \leq 4r^2, \tag{3.1}$$

$$\|F_r(u) - F_r(v)\|_{L^2L^3} \leq C_\theta r \|u - v\|_{E_T}, \qquad C_\theta = \max\{8, 6 + 9L_\theta\}. \tag{3.2}$$

These bounds are pathwise, and F_r preserves progressive measurability.

Proof. The support of the cutoff lies before A_u reaches $2r$, where the E norm is at most $2r$; Holder's inequality gives (3.1). Put $d = \|u - v\|_{E_T}$. If $d \geq r$, the sum of the two bounds is $8r^2 \leq 8rd$. If $d < r$, then on the union J_* of the cutoff supports both cumulative norms are less than $3r$. Hence $\|\mathbf{1}_{J_*} u\|_{E_T}, \|\mathbf{1}_{J_*} v\|_{E_T} \leq 3r$. With $\theta_u(t) = \theta(A_u(t)/r)$, decompose

$$F_r(u) - F_r(v) = \theta_u(u \otimes u - v \otimes v) + (\theta_u - \theta_v) v \otimes v.$$

The two L^2L^3 bounds are $6rd$ and $9L_\theta rd$, respectively. Outside J_* the difference is zero. This proves (3.2). The measurability claim follows from adaptedness and continuity of A_u. □

3.2 Construction and positive lifetime

Use $T = 1$ and abbreviate $M = M_1$ and $C_0 = C_0(1)$. Enlarge the positive constants if necessary and choose

$$r = (4MC_DC_\theta)^{-1}. \tag{3.3}$$

For deterministic $f \in X$ consider the map on the whole space $\mathcal{E}_1$

$$\Phi_f(u) = R_1\big[S(\cdot)f - DF_r(u)\big].$$

Lemmas 2.1, 2.3 and 3.1 imply

$$\|\Phi_f(u) - \Phi_f(v)\|_{\mathcal{E}_1} \le MC_DC_\theta r\|u - v\|_{\mathcal{E}_1} = \tfrac{1}{4}\|u - v\|_{\mathcal{E}_1}.$$

The map takes $\mathcal{E}_1$ into itself. In particular, the pathwise bound (3.1) gives the necessary fourth probability moment of the truncated forcing. Picard iteration converges in this Banach space to a unique fixed point U_f, for every size of f. Its equation is

$$U_f = S(\cdot)f - DF_r(U_f) + KU_f. \tag{3.4}$$

Equations (2.3), (2.4) and (2.11) supply an adapted continuous X version with $U_f(0) = f$. It agrees with the progressive L^6 representative almost everywhere. Separability of X and the usual completion of the filtration permit one common null-set modification for the continuous identity.

Define

$$\tau_f = \inf\{t \in [0,1] : A_{U_f}(t) \ge r\} \wedge 1, \qquad \inf \varnothing = 1. \tag{3.5}$$

Continuity and adaptedness make this a stopping time, and $A_{U_f}(0) = 0$ implies $\tau_f > 0$ almost surely. On $[0, \tau_f]$ the cutoff equals one. Thus $u_f = U_f|_{[0,\tau_f]}$ solves (1.3) with paths in $C([0,\tau_f]; X) \cap E_{\tau_f}$. Only positive-time L^6 values are used in the quadratic term; no L^6 initial datum is required.

3.3 Uniqueness in the whole stated class

Let u, v be any two solutions in the class of Theorem 1.1, with the same initial value and Brownian motions. Fix an integer $m \ge 1$ and put $\gamma = \tau_u \wedge \tau_v \wedge m$. Set $\varepsilon = (4M_mC_D)^{-1}$. Starting at $s_0 = 0$, define

$$s_{j+1} = \inf\left\{t \ge s_j : \int_{s_j}^{t} (\|u(a)\|_6 + \|v(a)\|_6)^4\, da \ge \varepsilon^4\right\} \wedge \gamma.$$

The norm integrands are zero after γ for this definition. These are stopping times because the accumulated adapted integrals are continuous. Every completed threshold consumes ε^4 of a finite pathwise integral, so almost every path reaches γ in finitely many slabs.

Assume inductively that $u = v$ through s_j. On $[0, s_{j+1}]$ put $w = u - v$; it is zero before s_j. Extend the tensor integrand

$$F = \mathbf{1}_{(s_j, s_{j+1}]}(u \otimes u - v \otimes v)$$

by zero outside its slab and put $v_0 = -DF$. Subtracting the mild equations gives $w - Kw = v_0$ up to s_{j+1}. By (2.2) and Holder,

$$\|v_0\|_{E_{s_{j+1}}} \le 2C_D\varepsilon\|w\|_{E_{s_{j+1}}}. \tag{3.6}$$

The stopped norm of w is at most 2ε, since its past is zero. Thus w and v_0 have finite fourth probability moments on this stopped interval, although the original solutions have no assumed moments. Apply (2.13) and then (3.6):

$$\|w\|_{L^4(\Omega;E_{s_{j+1}})} \le 2M_m C_D \varepsilon \|w\|_{L^4(\Omega;E_{s_{j+1}})} = \tfrac{1}{2}\|w\|_{L^4(\Omega;E_{s_{j+1}})}.$$

Hence $w = 0$ almost everywhere, and continuity in X gives equality at every time through s_{j+1}. A countable intersection over j and then m proves equality on the entire common lifetime. The estimate starts at time zero with an already vanishing past; it uses no random restart theorem.

3.4 Continuous dependence on common local intervals

The same θ, r work for every deterministic datum. The fixed-point estimate gives

$$\|U_f - U_g\|_{\mathcal{E}_1} \le MC_0\|f - g\|_X + \tfrac{1}{4}\|U_f - U_g\|_{\mathcal{E}_1}. \tag{3.7}$$

In particular, absorbing the last term yields $\|U_f - U_g\|_{\mathcal{E}_1} \le 2MC_0\|f - g\|_X$. Subtracting (3.4) and using the continuous-trace estimates for S, D, K and (3.2) then gives

$$\|U_f - U_g\|_{L^4(\Omega;C([0,1];X))} \le C\|f - g\|_X. \tag{3.8}$$

Suppose $f_n \to f$ in X. Restriction of these estimates to $[0, \tau_{f_n} \wedge \tau_f]$ proves fourth-moment convergence in the sum of the two norms asserted in the theorem, hence convergence in probability. To show that the common intervals do not collapse, let

$$\rho = \inf\{t \in [0,1] : A_{U_f}(t) \ge r/2\} \wedge 1.$$

Then $0 < \rho \le \tau_f$ almost surely. If $\tau_{f_n} < \rho$, continuity gives $A_{U_{f_n}}(\tau_{f_n}) = r$ and $A_{U_f}(\tau_{f_n}) < r/2$. The reverse triangle inequality implies $\|U_{f_n} - U_f\|_{E_1} \ge r/2$, and therefore

$$\Pr(\tau_{f_n} < \rho) \le 16 r^{-4}\mathbb{E}\|U_{f_n} - U_f\|_{E_1}^4 \longrightarrow 0.$$

This proves the stability assertion and completes the proof of Theorem 1.1.

AI-use declaration

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